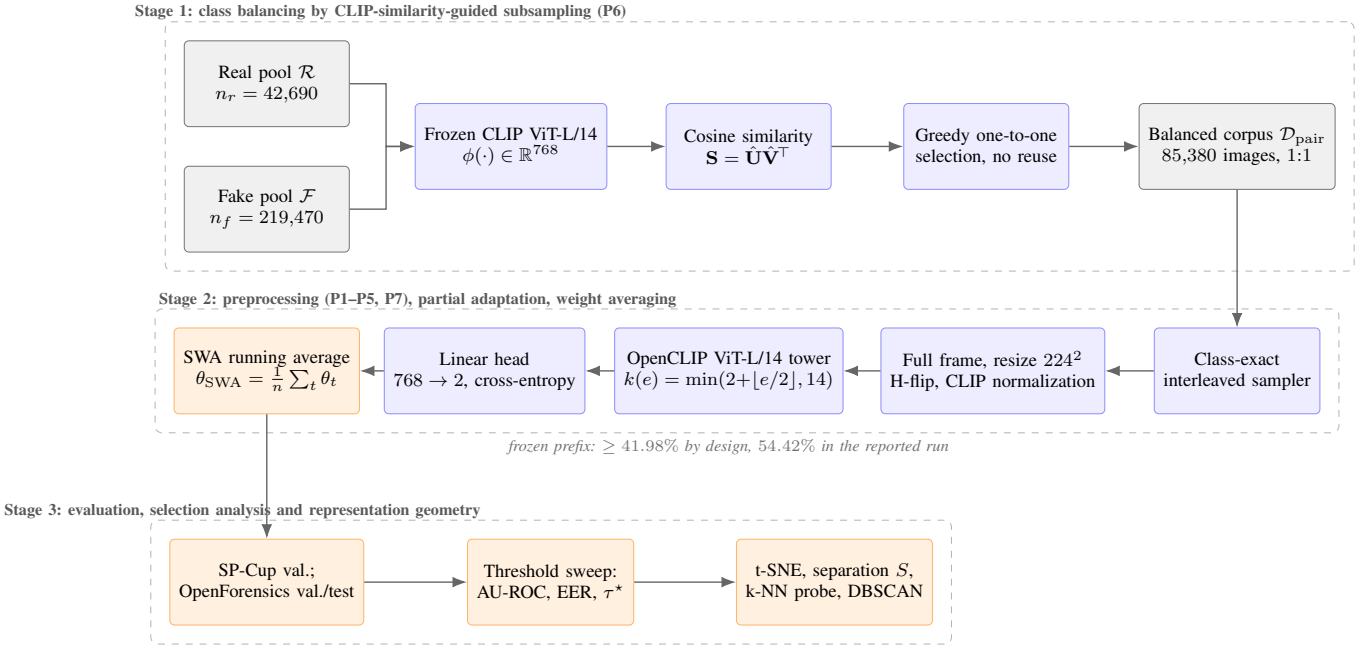

Stage 1: class balancing by CLIP-similarity-guided subsampling (P6)
Real pool $\mathcal{R}$
$n_r = 42{,}690$
Fake pool $\mathcal{F}$
$n_f = 219{,}470$
Frozen CLIP ViT-L/14
$\phi(\cdot) \in \mathbb{R}^{768}$
Cosine similarity
$\mathbf{S} = \hat{\mathbf{U}}\hat{\mathbf{V}}^\top$
Greedy one-to-one
selection, no reuse
Balanced corpus $\mathcal{D}_{\text{pair}}$
85,380 images, 1:1
Stage 2: preprocessing (P1–P5, P7), partial adaptation, weight averaging
Class-exact
interleaved sampler
Full frame, resize 224^2
H-flip, CLIP normalization
OpenCLIP ViT-L/14 tower
$k(e) = \min(2{+}\lfloor e/2 \rfloor, 14)$
Linear head
$768 \to 2$, cross-entropy
SWA running average
$\theta_{\text{SWA}} = \frac{1}{n}\sum_t \theta_t$
frozen prefix: $\geq 41.98\%$ by design, 54.42% in the reported run
Stage 3: evaluation, selection analysis and representation geometry
SP-Cup val.;
OpenForensics val./test
Threshold sweep:
AU-ROC, EER, $\tau^\star$
t-SNE, separation S,
k-NN probe, DBSCAN